



Autacoids

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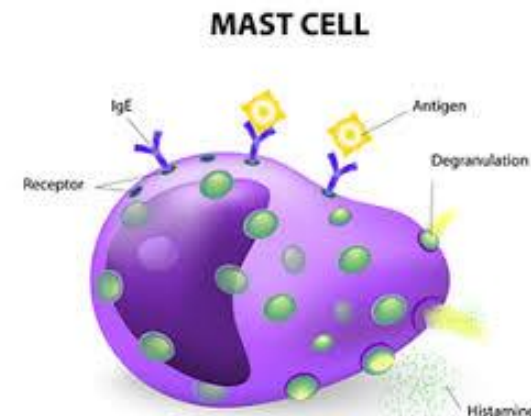
Objectives

1. Describe the synthesis, distribution, and functions of histamine and serotonin.
2. Explain the types of histamine and serotonin receptors and their effects.
3. List the main drugs acting on histamine and serotonin pathways.
4. Compare different types of antihistamines based on their properties and effects.
5. Apply the use of antihistamines and serotonin-related drugs in common clinical conditions.
6. Recognize important adverse effects and contraindications of these drugs.

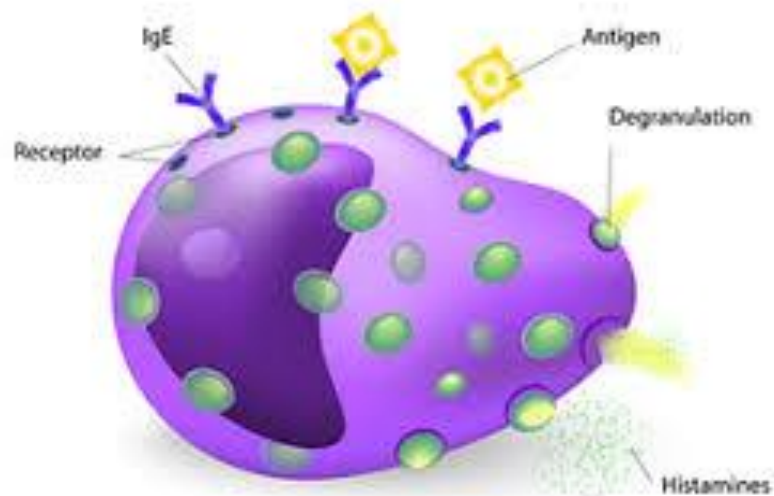


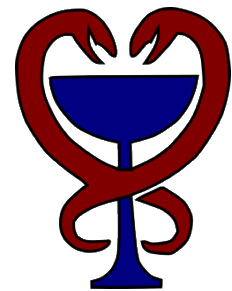
Autacoids

- Autos = self; Akos = cure (Greek).
- Naturally occurring substances.
- Differ from hormones and neurotransmitters.
- Sites of action restricted to the synthesis area (local effect).
- Short duration of action.
- They include prostaglandins (PGs), histamine, and serotonin(5 HT).



Histamine

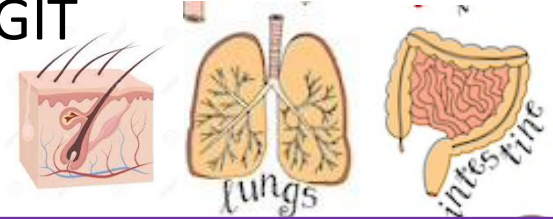




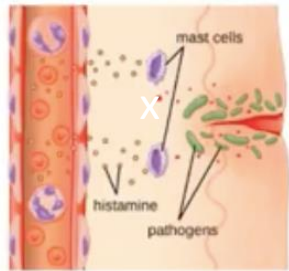
Drugs & Foreign chemicals



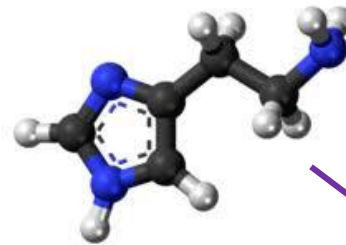
Present in all tissue especially lung, skin and GIT



Tissue injury



Secreted in response to:

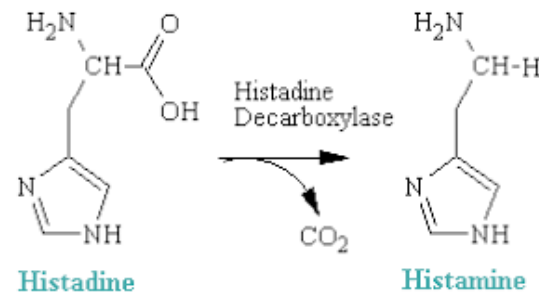
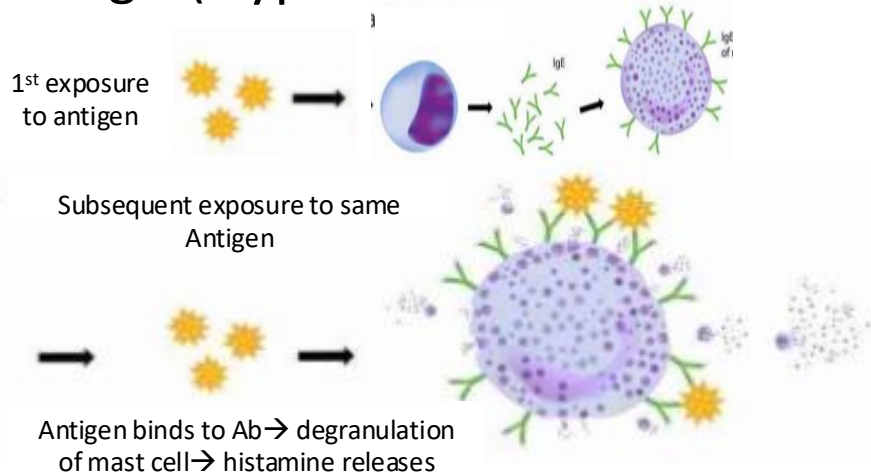


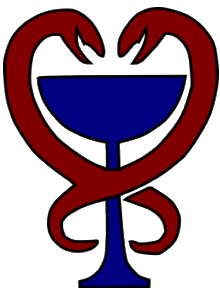
Histamine

Synthesis and secreted from: mast cells in tissue and basophil in blood

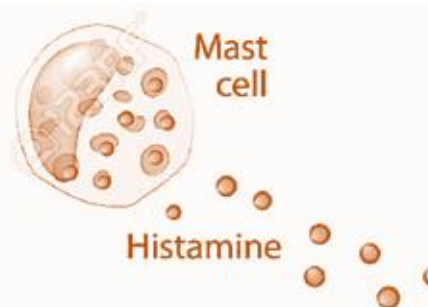


Allergic (Hypersensitivity) reaction





Histamine: Actions

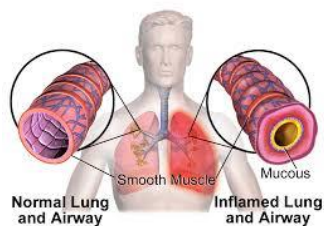


H₁

Vasodilation



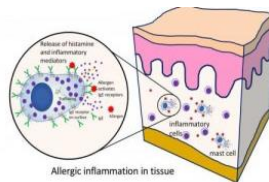
Bronchoconstriction



wakefulness

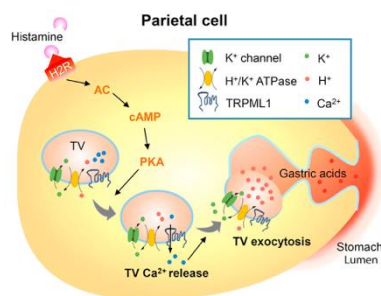


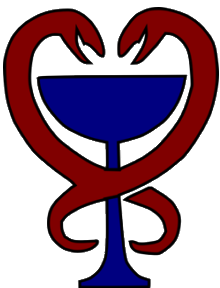
Itching



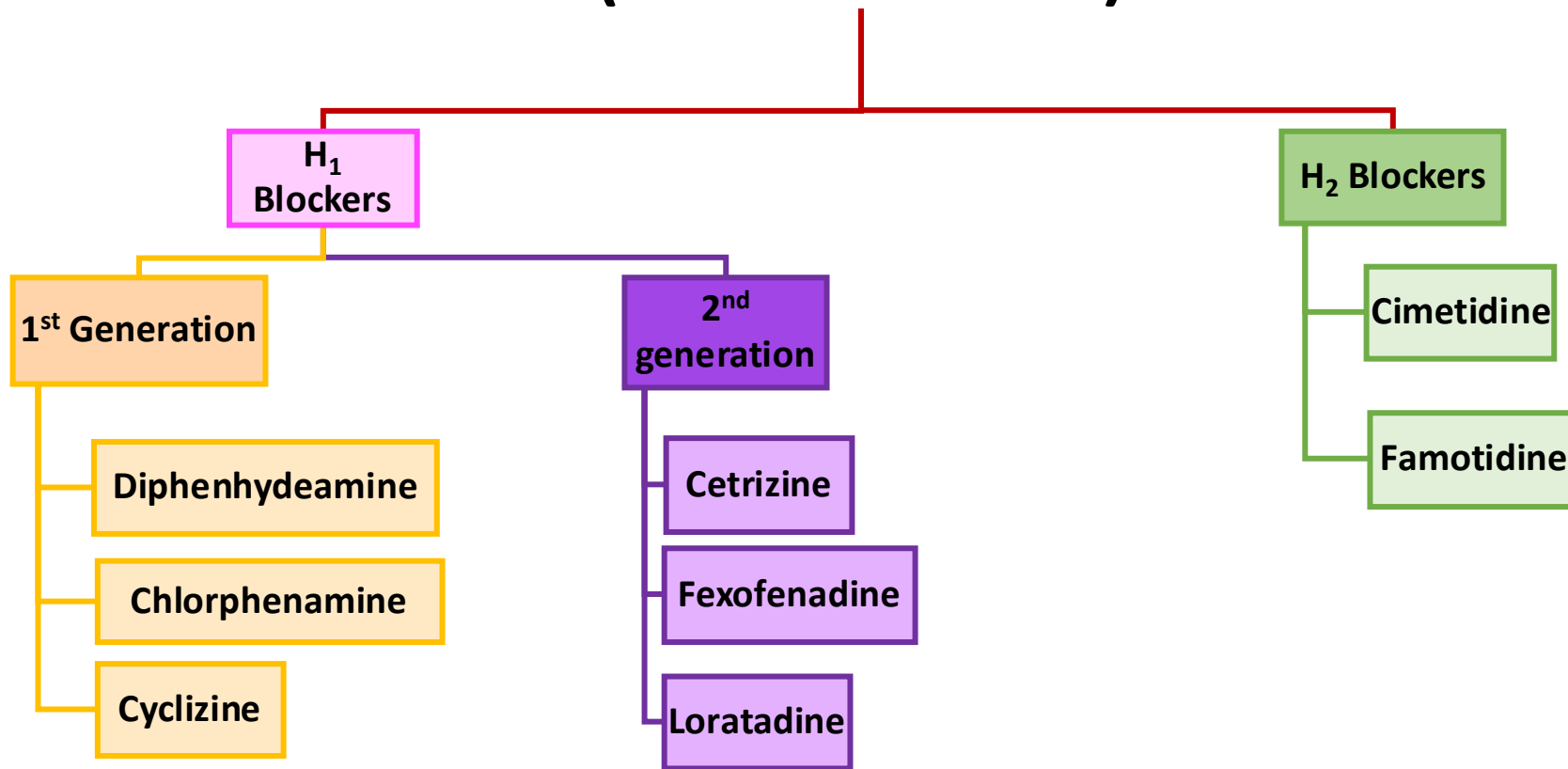
H₂

Increase gastric HCL secretion





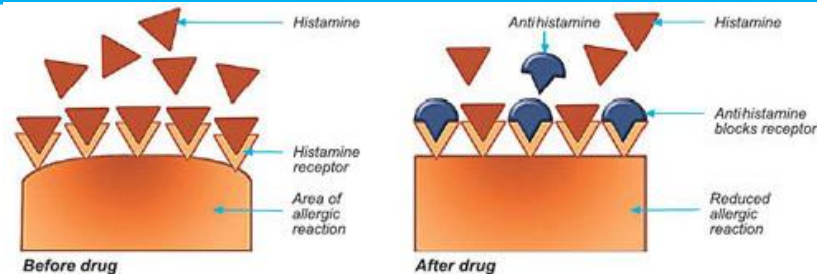
Histamine receptor blockers (anti-histamine)



1. H₁ blockers:

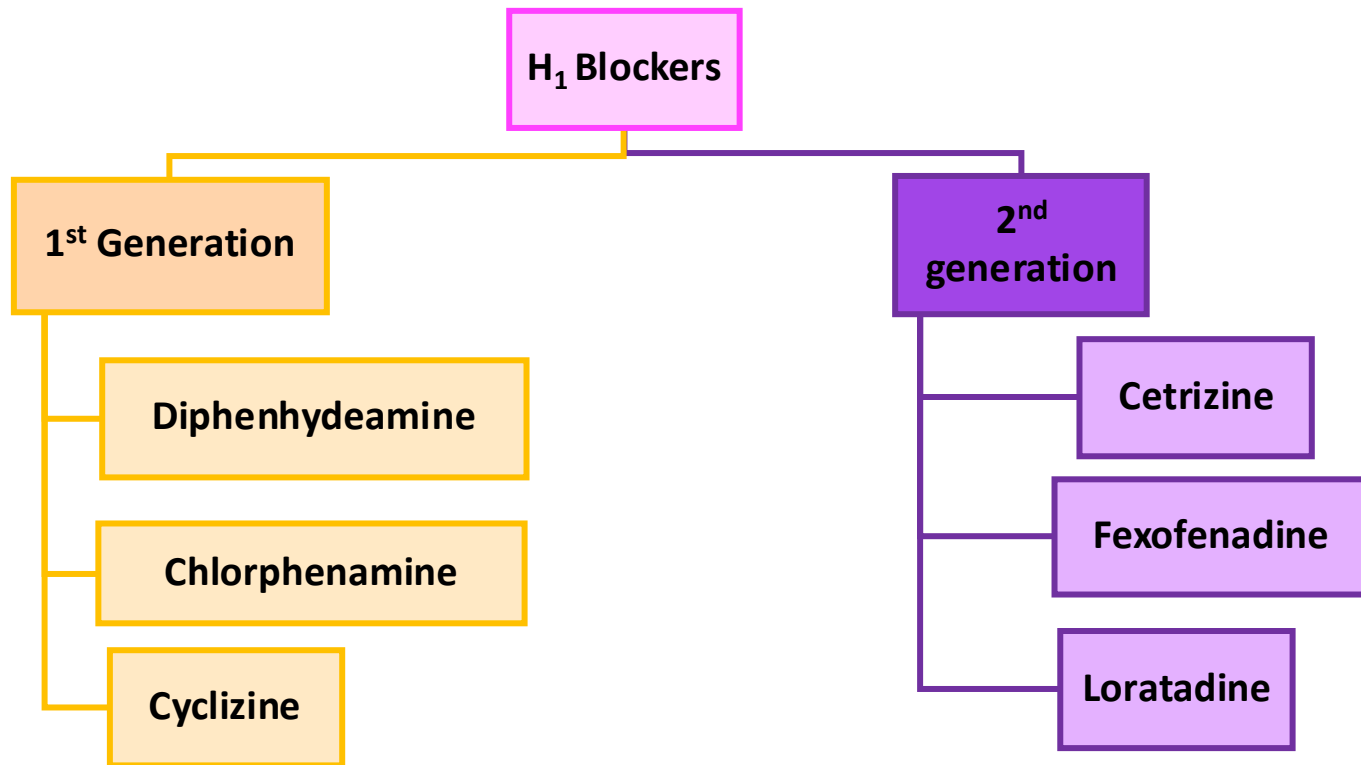
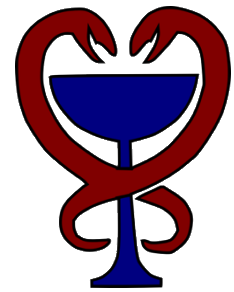
Mechanism of action

- competitively antagonize H1 receptor.
- Non-selectively block cholinergic and α -adrenergic receptors (**1st generation**).



Pharmacodynamics & Pharmacokinetics

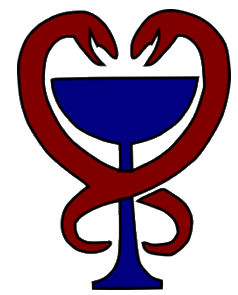
- Route of administration: Oral, topical, IV.
- Metabolized in liver.
- **1st generation** : Lipid soluble → Cross BBB → Sedative, anti-emetic effects.
- **2nd generation** : Less lipid soluble → NOT cross BBB → NOT Sedative.
- 2nd generation has longer effect.



- Cross BBB
- Potentially cause S/E
- Cause sedation
- Nonselective : anticholinergic and antiadrenergic
- 3-4 t/ day



- Does not cross BBB
- Less S/E (safer)
- No sedation
- More selective to H₁
- 1-2 t/ day



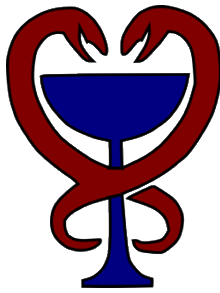
1. H₁ blockers:

Therapeutic uses:

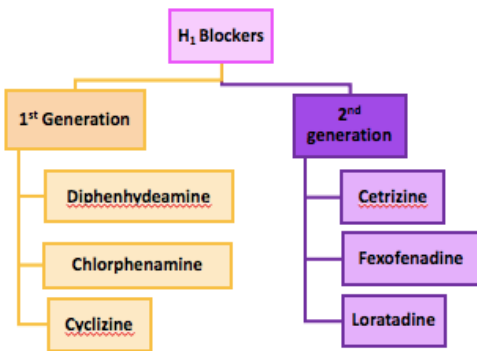
- **Treatment of allergic conditions** (hay fever, urticaria, atopic rhinitis). **2nd generation is preferred**

- **Sleeping pills.** (diphenhydramine)
- **Anti-motion sickness** (diphenhydramine, cyclizine).
- **Pregnancy -induced vomiting** (diphenhydramine)
- **Antiparkinsonism:** diphenhydramine (early stages)





1. H₁ blockers:



Adverse effects

- Sedation.
- Dry mouth and blurred vision, tachycardia, tremor (antimuscarinic effect)
- Urinary retention (antimuscarinic effect)
- Orthostatic hypotension (a- adrenergic blocker effect).



[illegible]

Loratadine “Low” “Jeans”

Cetirizine
"Citrus" "Magazine"

Cyclizine

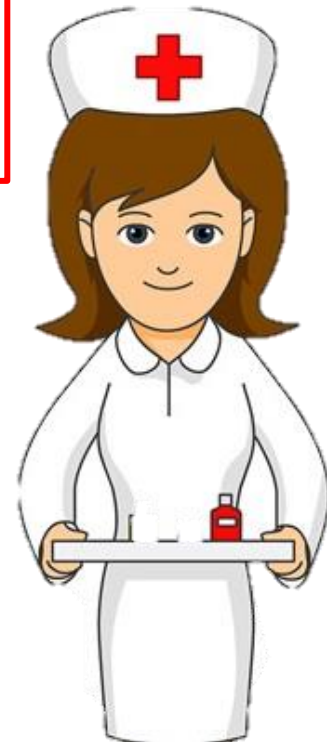
“Cycle” “magazine”

Diphenhydramine

1st generation
2nd generation

Clinical counselling points

- Avoid driving (sedation)
- Take before travel (motion sickness)
- Stop before allergy testing
- Anticholinergic effects (dry mouth, urinary retention)



H₂ Blockers

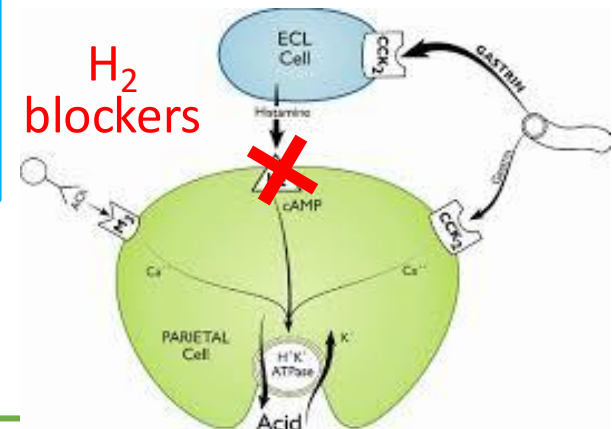
Cimetidine

Famotidine

3. H₂ blockers:

Mechanism of action

selectively competitive antagonize H₂ receptor on parietal cells → decrease gastric HCL production.



Pharmacodynamics & Pharmacokinetics

- Route of administration: Oral, IV (OTC medications).
- **Cimetidine:** has **short duration**. It can **inhibit the metabolism** of many drugs and **has anti-androgenic & prolactin-stimulating** effects.
- **Famotidine:** **more** potent than cimetidine with **longer** action. Unlike cimetidine it is **not** an enzyme inhibitor, **no** anti-androgenic or prolactin-stimulating effects.



Therapeutic uses:

- Peptic ulcer

H₂ Blockers

Cimetidine

Famotidine

3. H₂ blockers:

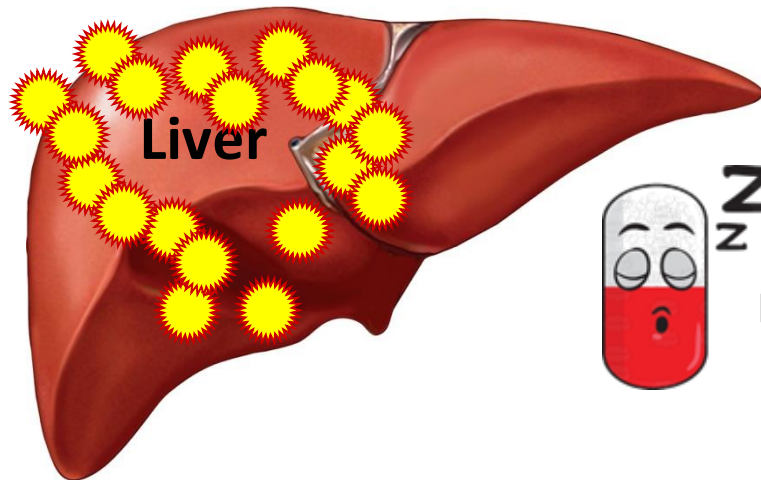
Adverse effects

- **Cimetidine** has antiandrogen effects. Gynecomastia in male and irregular menstruation in female.
- Drug drug interactions: Cimetidine is a potent **inhibitor** of hepatic drug-metabolizing enzymes (P-450) → risk of drug toxicity such as warfarin

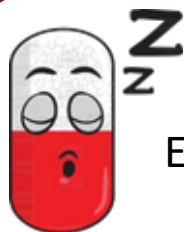


Drug X

Metabolized by liver

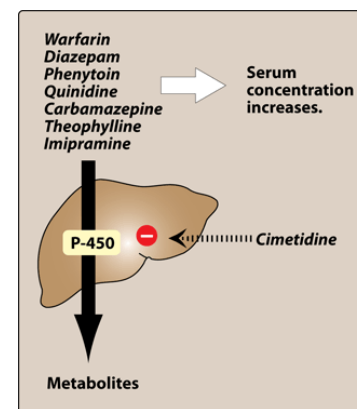


Liver



Cimetidine

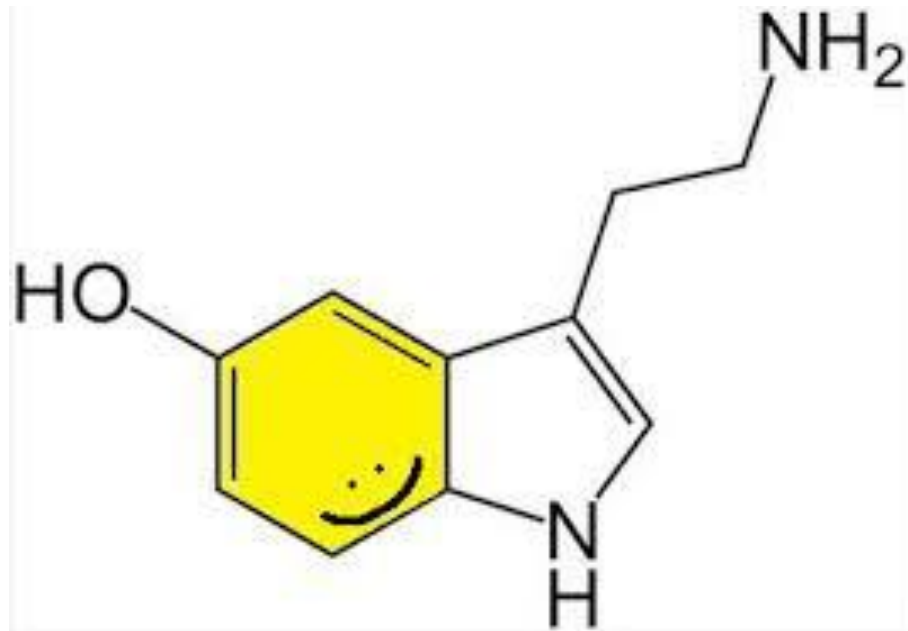
Enzyme Inhibitor

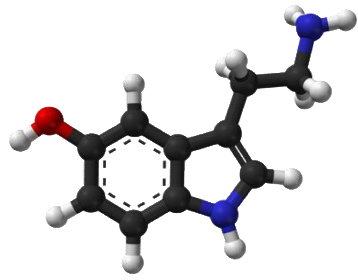


This increases the amount of Drug X in the body and could lead to an overdose or toxic effects

Serotonin

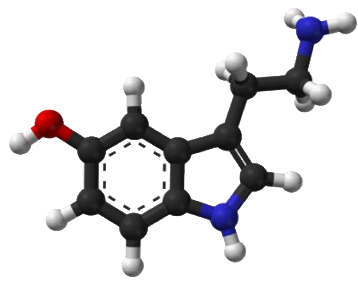
5-HT



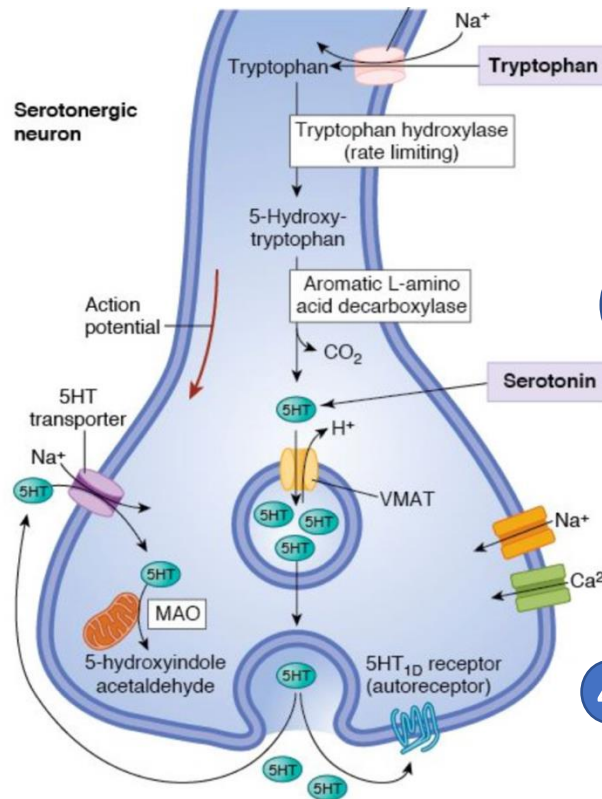


Serotonin (5-HT) biosynthesis

- Present in GIT - enterochromaffin cells (90%), platelets and brain.
- Synthesized from tryptophan (in diet) in two steps.
- Cell storage in granules similar to catecholamines.



Serotonin (5-HT) biosynthesis



1 -Produced from tryptophan.

2 Stored in the vesicles in enterochromaffin cells in gut and neurons.

3 - Metabolized by MAO..

4 Partially taken back by serotonin reuptake transporter.

5-HT_{1A}

anxiety/depression

5-HT_{1D}

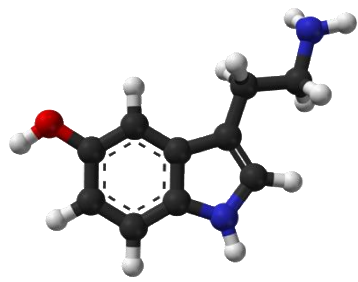
migraine

5-HT₂

CNS
CVS

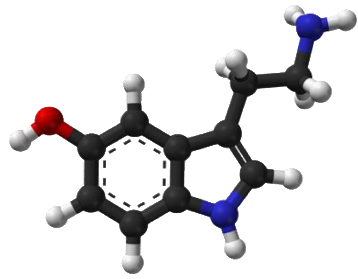
5-HT₃

CNS →
induce vomiting



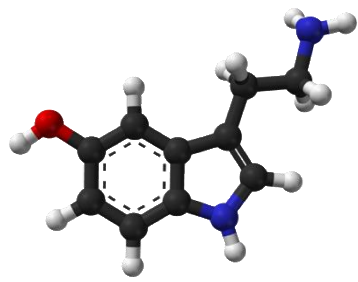
Serotonin (5-HT) biosynthesis

- At least 15 types and subtypes.
- 5HT-1A: role in anxiety/depression.
- 5HT-1D: role in migraine.
- 5HT-2: role in CNS various behaviors, and in cardiovascular system.
- 5-HT3: role in nausea and vomiting esp. due to Chemotherapy.



Serotonin (5-HT) biosynthesis

- Central neurotransmitter.
- GI tract: uncertain; motility?
- In carcinoid tumors: large amounts released leading to diarrhea, bronchoconstriction and edema.
- Platelets: 5-HT₂ receptors → aggregation and vasoconstriction.



Serotonin (5-HT) agonists

5-HT₁

Sumatriptan

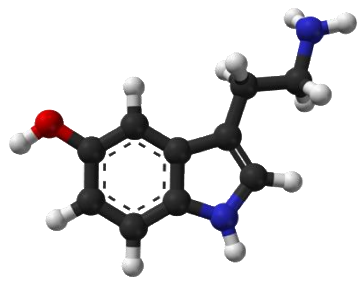
-1st line tx of acute **migraine** and cluster headache (relief: 10 to 15 minutes following nasal spray) .

-Selective serotonin-receptor agonist (short duration of action)

-Not Recommended For Patients With Coronary Vascular Disease Risk.
-**Formulations:** subcutaneous injection, oral, nasal spray.



Sumatriptan



Serotonin (5-HT) antagonists

5-HT₃ Antagonist **Ondansetron**

5-HT₃ in CNS → induce vomiting

Control chemotherapy-induced vomiting and post-op vomiting (CINV)

Ondansetron → QT prolongation

